ENVISION-X 2.0

NATIONAL LEVEL CODING CHALLENGE 2026

TITLE PAGE

Team Name-Team Nexis

Problem Statement Title- **Companies struggle to detect fake candidates, scripted answers and cheating in interviews because traditional hiring relies on subjective human judgment.**

Theme-Future-Ready Products & Services

Team Leader Name-Chirag Keswani

University Name-Amity University Chhattisgarh

IDEA TITLE

AI Interview Truth Analyzer

An intelligent AI system that analyzes voice, facial expressions, eye movement, and behavioral patterns during interviews to detect fake confidence, scripted answers, and cheating.

The system generates a Truth Score and Authenticity Report to help recruiters make accurate hiring decisions.

How the System Works

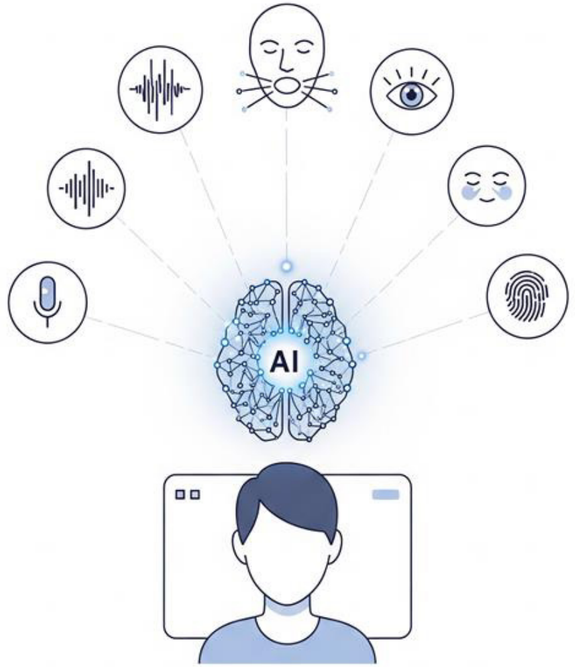
- Voice stress patterns detect nervousness or deception
- Facial expressions analyzed using computer vision
- Eye movement tracking detects suspicious behavior
- Speech patterns identify scripted answers
- Behavioral confidence vs answer correctness

Output

- Truth Score (0–100)
- Confidence Score
- Risk of cheating detection
- Behavior analysis report

Innovation

- Combines Artificial Intelligence + Psychology + Recruitment Technology
- Focuses on interview authenticity detection



TECHNICAL APPROACH

Technologies Used:

- Python
- OpenCV – facial expression & eye tracking
- Speech Recognition API
- Machine Learning models
- Natural Language Processing (NLP)
- TensorFlow / PyTorch

Implementation Process

1. Candidate starts online interview
2. Camera and microphone capture behavior
3. AI analyzes facial expressions and voice tone
4. NLP analyzes answers
5. System calculates Truth & Confidence Score

FEASIBILITY AND VIABILITY

Feasibility

The project is feasible because required technologies like computer vision, speech analysis, and machine learning already exist.

Potential Challenges

* Privacy concerns
* Accuracy of deception detection
* Real-time processing requirements

Solution:

- Use secure data encryption
- Train AI models with large datasets
- Optimize algorithms for faster processing

IMPACTS AND BENEFITS

Impact on Recruiters

- Better hiring decisions
- Reduced hiring fraud
- Faster screening process

Social Benefits

- Promotes fair recruitment
- Reduces fake candidates in companies

Economic Benefits

- Saves company time and hiring costs

RESEARCH AND REFERENCES

- Research papers on AI-based emotion detection
- Studies on voice stress analysis
- Machine learning models for facial expression recognition

References:

- * IEEE Research Papers
- * Google AI Research
- * OpenCV Documentation